MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)

MG4DSCECT202 - IOT System Design

Duration: 1.5 hours **Maximum Marks: 50**

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A
Multiple Choice Questions
Answer All Questions. Each question carries 1 mark

1	Which layer of IoT is responsible for data collection using sensors? A. Application layer B. Cloud layer C. Network layer D. Perception layer CO1	R	1
2	Which sensor is commonly used for motion detection? A.LDR B. PIR C. IR D. Gas sensor	R	2
3	ESP8266 mainly provides which feature? A. Bluetooth only B. GSM communication C. Wi-Fi connectivity D. GPS tracking	R	2
4	MQTT follows which communication model? A. Client–Server B.Peer-to-Peer C. Publish–Subscribe D. Master–Slave	U	2
5	Which actuator works using compressed air? A. Hydraulic B. Mechanical C. Magnetic D. Pneumatic	R	2
6	ThingSpeak is mainly used for A. Device manufacturing B. Cloud data storage and visualization C. Sensor fabrication D. PCB design	U	2
7	Medical IoT is mainly used for A. Industrial automation B. Health monitoring C. Smart traffic D. Power management	U	3
8	Which is a major challenge in IoT systems? A. High bandwidth B. Privacy issues C. Unlimited storage D. Low latency	R	3

9	Smart wearable devices belong to	U	3
	A. Industrial IoT B. Agricultural IoT		
	C. Consumer IoT D. Commercial IoT		
10	Drone delivery systems are an example of	U	3
	A. Cloud computing B. AI in IoT		
	C. Network security D. Sensor fusion		

1*10=10

Part B

Short Answer Questions

Answer 4 Questions. Each question carries 5 marks

11	Explain the building blocks of IoT with a neat block diagram.	U	1
12	Describe the layers of IoT architecture and their functions.	U	1
13	Explain the working principle and applications of PIR and Ultrasonic sensors.	U	2
14	Write a short note on ESP32 and Raspberry Pi Zero highlighting their features.	R	2
15	Explain MQTT protocol and its publish–subscribe mechanism.	U	2
16	Discuss privacy and legal challenges in IoT systems.	A	3

5*4=20

Part B

Short Answer Questions

Answer 4 Questions. Each question carries 5 marks

17	a) Explain different types of IoT classification with suitable examples. b) Describe applications of IoT in smart cities and agriculture.	U A	3
18	a) Explain various IoT communication protocols: MQTT, CoAP, XMPP and DDS. b) Compare Blynk, ThingSpeak, Arduino Cloud and Firebase as IoT platforms.	A	2
19	a) Discuss IoT applications in healthcare and lifestyle monitoring. b) Explain the role of AI in IoT with examples such as automated vehicles and smart wearables.	A U	3
20	a) Explain IoT based home automation and security systems. b) Discuss ethical and environmental challenges associated with IoT deployment.	A	3

10*2=20